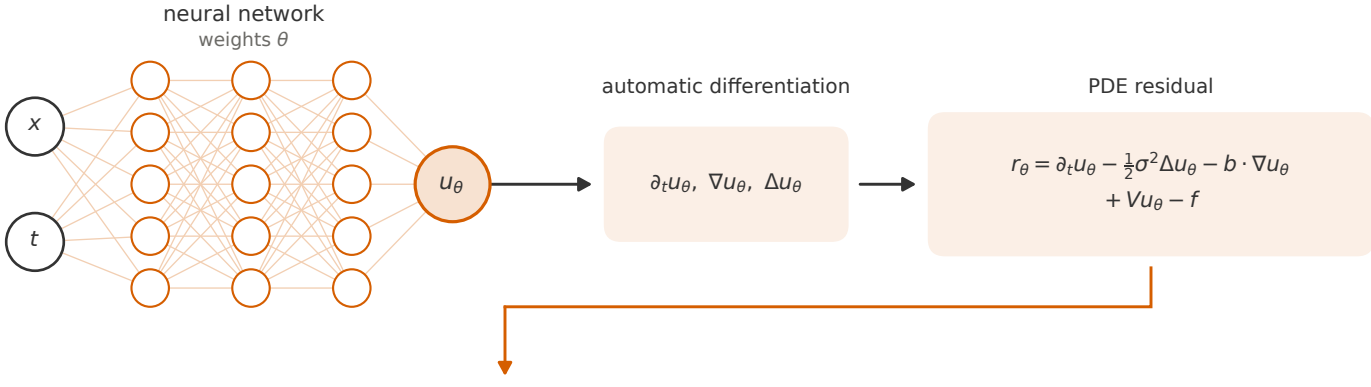
neural network
weights θ
x
t
u_θ
automatic differentiation
$\partial_t u_\theta,\ \nabla u_\theta,\ \Delta u_\theta$
PDE residual
$r_\theta = \partial_t u_\theta - \frac{1}{2}\sigma^2 \Delta u_\theta - b \cdot \nabla u_\theta + V u_\theta - f$
$\mathcal{L}(\theta) = \mathcal{L}_{\text{res}} + \lambda_{\text{FK}}\,\mathcal{L}_{\text{FK}} + \lambda_?\,\mathcal{L}_?$
$\frac{1}{N}\sum_i r_\theta(x_i, t_i)^2$
PDE residual
$\frac{1}{N_{\text{FK}}}\sum_k (u_\theta(x_k, t_k) - \bar{u}_k)^2$
Feynman–Kac targets